

Question 1 (10 marks)**Marks**

a) For the following functions,

8

i) Sketch the following showing all the important features.

ii) Find the natural domain and range.

A) $y = |x+1| - 1$

B) $y = \frac{2}{x-3}$

b) A function is defined by $f(x) = \begin{cases} x^2 & \text{for } x \leq -1 \\ x+1 & \text{for } x > -1 \end{cases}$ Evaluate $f(-2) + 2f(0) - f(3)$.**2****Question 2** (10 marks)**Start this question in a new booklet.**

a) Show by 'completing the square' that

4 $x^2 + 4x + y^2 - 2y - 4 = 0$ is a circle, and state its centre and radius.b) A function is defined as $f(x) = x^3 - 5x$ **2**

Show this function is either even, odd or neither.

c) Find the exact value of (i) $\tan 240^\circ$ **1**(ii) $\sin(-45)^\circ$ **1**d) Given that $\tan \theta = \frac{-5}{12}$ and $\sin \theta > 0$, find**2**(i) $\sin \theta$ (ii) $\sec \theta$ *continued on the next page*

Question 3 (10 marks) **Start this question in a new booklet.** **Marks**

- a) If $\sin \theta = \cos 35^\circ$ find θ , if θ is acute. **1**
- b) Simplify the following $\cot \theta - \cot \theta \cos^2 \theta$ **2**
- c) Prove $(1 + \tan^2 A) \cos^2 A = 1$ **2**
- d) Solve the following, for $0^\circ \leq \theta \leq 360^\circ$
- (i) $\tan \theta = \frac{1}{\sqrt{3}}$ **2**
- (iii) $2 \cos 2\theta - 1 = 0$ **3**

Question 4 (9 marks) **Start this question in a new booklet.**

- a) Prove $\frac{1}{1 + \sin x} + \frac{1}{1 - \sin x} = 2 \sec^2 x$ **3**
- b) From a harbour H, a boat B travelled 20km on a bearing of 125 degrees .
At the same time a yacht Y sailed 25km, on a bearing of 245 degrees.
- (i) Draw a diagram, marking on it all the information supplied. **1**
- (ii) Find the distance of the boat B, from the yacht Y.
Give your answer as an exact value. **2**
- (ii) Find the bearing of the yacht Y from the boat B.
Give your answer to the nearest degree. **3**

continued on the next page

Question 5 (10marks) **Start this question on a new page.** **Marks**

The points A, B and C have the co-ordinates $(-2,2)$, $(-1, -5)$ and $(6,-6)$ respectively.

- | | |
|---|----------|
| a) Sketch the triangle ABC, on a number plane using a ruler. Make the diagram about $\frac{1}{2}$ page in size. | 1 |
| b) Show the midpoint P of AC has the co-ordinates $(2,-2)$. | 1 |
| c) Calculate the gradient of AC. | 1 |
| d) Find the exact value of the length of side AC. | 1 |
| e) Show AC and BP are perpendicular. | 2 |
| f) Find the area of triangle ABC. | 2 |
| g) Find the equation of the line AC. | 2 |

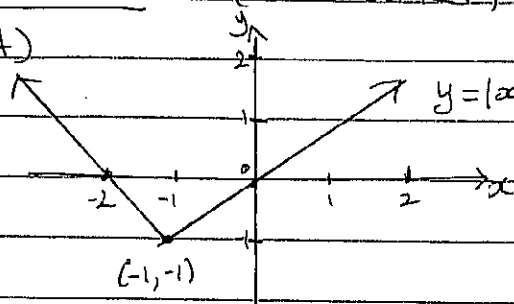
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Year 11 2013 Mathematics Assessment 2

Question 1 (10 marks)

a) (A)

(1)



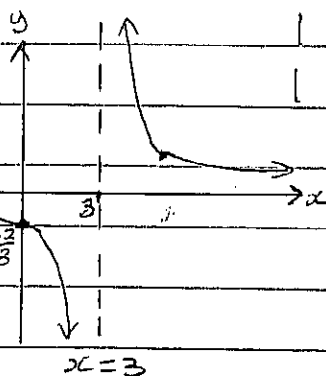
1 correct shape

1 all points correct

(ii) Domain: all real x ✓ 1 mark each

Range: $y \geq -1$ ✓

(B) $y = \frac{2}{x-3}$



1 correct shape

1 correct

(ii) Domain: all real $x, x \neq 3$ ✓

Range: all real $y, y \neq 0$ ✓

1 mark each.

y int
and both
asymptotes

b) $f(x) = \begin{cases} x^2, & \text{for } x \leq -1 \\ x+1, & \text{for } x > -1 \end{cases}$

$$f(-2) = -2^2$$

$$= 4$$

$$f(0) = 0+1$$

$$= 1$$

$$f(3) = 3+1$$

$$= 4$$

$$\therefore f(-2) + 2f(0) - f(3) = 4 + 2 \times 1 - 4 \quad \checkmark \quad 2 \text{ marks.}$$

$$= 2 \quad \checkmark$$

2013 Year 11 Mathematics Assessment 2

Question 2 (10 marks)

a) $x^2 + 4x + y^2 - 2y - 4 = 0$ (4 marks)

$$x^2 + 4x + \left(\frac{4}{2}\right)^2 + y^2 - 2y + \left(\frac{1}{2}\right)^2 = 4 + 5 \quad \checkmark$$

$$(x+2)^2 + (y-1)^2 = 9 \quad \checkmark$$

$\therefore$ its a circle centre $(-2, 1)$ $\checkmark$

and radius 3 $\checkmark$

b) $f(x) = x^3 - 5x$

$$f(-x) = (-x)^3 - 5(-x) \quad \checkmark$$

$$= -x^3 + 5x$$

1 mark

$$-f(x) = -(x^3 - 5x)$$

= 2 marks.

$$= -x^3 + 5x$$

$$\therefore f(-x) = -f(x)$$

$\therefore f(x)$ is an odd function $\checkmark$

1 mark

c) (i) $\tan 240^\circ = +\tan 60^\circ$

$$= +\sqrt{3}$$

$\checkmark$

1 mark

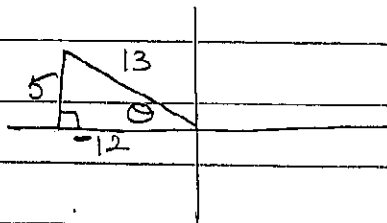
(ii) $\sin(-45^\circ) = -\sin 45^\circ$

$$= -\frac{1}{\sqrt{2}}$$

$\checkmark$

1 mark

d) $\tan \theta = -\frac{5}{12}$, $\sin \theta > 0$



(i) $\sin \theta = \frac{5}{13}$ $\checkmark$ 1 mark

(ii) $\cos \theta = -\frac{12}{13}$

$\sec \theta = -\frac{13}{12}$ $\checkmark$ 1 mark.

2013 Year 11 Mathematics Ass 2

Question 3 (10 marks)

a) $\sin \theta = \cos 35^\circ$

$\therefore \theta + 35 = 90^\circ$

$\theta = 90 - 35^\circ$

$\theta = 55^\circ \quad \checkmark \quad 1 \text{ mark}$

b) $\cot \theta - \cot \theta \cos^2 \theta$

$= \cot \theta (1 - \cos^2 \theta)$

$= \cot \theta (\sin^2 \theta)$

$= \frac{\cos \theta}{\sin \theta} \times \sin^2 \theta$

$= \cos \theta \sin \theta \quad \checkmark \quad 2 \text{ marks}$

c) Prove $(1 + \tan^2 A) \cos^2 A = 1$

L.H.S. $= (1 + \tan^2 A) \cos^2 A$

$= \sec^2 A \times \cos^2 A$

2 marks.

$= \frac{1}{\cos^2 A} \times \cos^2 A$

$= 1$

$= \text{RHS.}$

$\therefore \text{LHS} = \text{RHS}$

$\therefore (1 + \tan^2 A) \cos^2 A = 1$

d) (i) $\tan \theta = \frac{1}{\sqrt{3}}$

$0^\circ \leq \theta \leq 360^\circ$

$\tan$ is positive in $\text{Q1} + \text{Q3}$

related angle

$\tan \theta = \frac{1}{\sqrt{3}}$

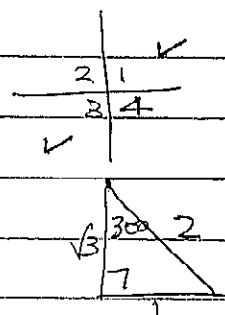
$\theta = 30^\circ \quad \checkmark$

$\therefore \text{Q1 } \theta = 30^\circ$

$\text{Q3 } \theta = 180 + 30^\circ = 210^\circ$

$\therefore \theta = 30^\circ, 210^\circ \quad \checkmark$

2 marks.



$$(d) (ii) 2\cos 2\theta - 1 = 0 \quad 0 \leq \theta \leq 360^\circ$$

$$\cos 2\theta = \frac{1}{2}$$

$$\text{let } x = 2\theta$$

$$0 \leq 2\theta \leq 720^\circ$$

$$\cos x = \frac{1}{2} \quad \therefore 0 \leq x \leq 720^\circ$$

$\cos$ is pos in $Q1 + Q4, Q5 + Q8$

related angle

$$\cos x = \frac{1}{2}$$

$$x = 60^\circ \checkmark$$

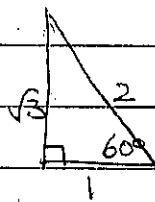
$$\therefore Q1 \quad x = 60^\circ$$

$$Q4 \quad x = 360 - 60 \\ = 300^\circ$$

$$Q5 \quad x = Q1 + 360^\circ \\ = 360^\circ + 60^\circ \\ = 420^\circ$$

$$Q8 \quad x = Q4 + 360^\circ \\ = 300^\circ + 360^\circ \\ = 660^\circ$$

6	2	1	5	✓
	3	4		
7			8	✓



$$\therefore x = 60^\circ, 300^\circ, 420^\circ, 660^\circ$$

$$2\theta = 60^\circ, 300^\circ, 420^\circ, 660^\circ$$

$$\theta = \underbrace{30^\circ, 150^\circ}_{\checkmark}, \underbrace{210^\circ, 330^\circ}_{\checkmark}$$

3 marks

2013 Year 11 Mathematics Ass 2

Question 4 (9 marks)

a) Prove $\frac{1}{1+\sin x} + \frac{1}{1-\sin x} = 2\sec^2 x$

$$\text{LHS} = \frac{1}{1+\sin x} + \frac{1}{1-\sin x}$$

$$= \frac{1(1-\sin x)}{(1+\sin x)(1-\sin x)} + \frac{1(1+\sin x)}{(1-\sin x)(1+\sin x)} \quad \checkmark$$

$$= \frac{(1-\sin x) + (1+\sin x)}{(1+\sin x)(1-\sin x)}$$

$$= \frac{2}{1-\sin^2 x} \quad \checkmark$$

$$= \frac{2}{\cos^2 x} \quad \checkmark$$

$$= 2\sec^2 x$$

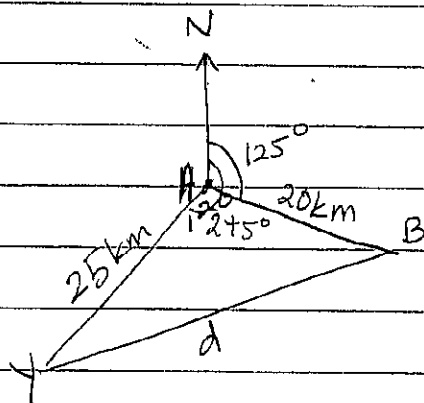
$$= \text{RHS.}$$

$$\therefore \text{LHS} = \text{RHS.}$$

$$\therefore \frac{1}{1+\sin x} + \frac{1}{1-\sin x} = 2\sec^2 x$$

3 marks.

b) (i)



$\checkmark$ 1 mark all info

$$(ii) d^2 = a^2 + b^2 - 2ab \cos D$$

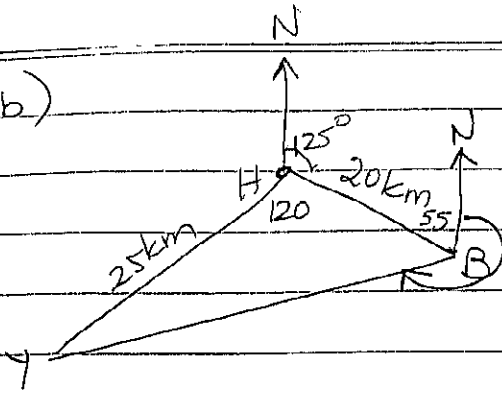
$$= 20^2 + 25^2 - 2 \times 20 \times 25 \times \cos 120^\circ \quad \checkmark$$

$$d^2 = 1525$$

$$\left. \begin{aligned} d &= \sqrt{1525} \\ d &= 5\sqrt{61} \end{aligned} \right\} \quad \checkmark$$

2 marks

Q4 (b)



$$(III) \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 120}{\sqrt{1525}} = \frac{\sin B}{25}$$

$$\sin B = \frac{25 \times \sin 120}{\sqrt{1525}}$$

$$= 0.5544159532$$

$$B = \sin^{-1}(0.5544159532)$$

$$B = 33.67049651$$

$$B = 34^\circ \text{ (n. degree)} \quad \checkmark$$

$$\angle NBH = 180^\circ - 125^\circ = 55^\circ \quad \checkmark$$

∴ the bearing of Y from B is

$$= 360^\circ - 55^\circ - 34^\circ$$

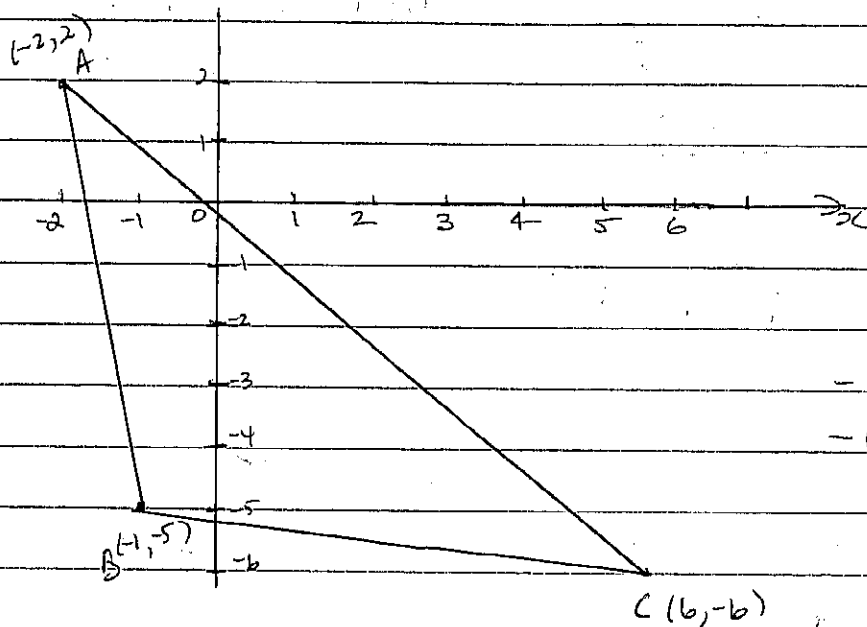
$$= 271^\circ \quad \checkmark$$

3 marks.

2013 Year 11 Mathematics Ass 2

Question 5 (10 marks)

(a)



1 mark.

- must use a ruler

- must be large...

(b) Midpoint of AC

$$P = \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right)$$

$$= \left(\frac{-2 + 6}{2}, \frac{2 + (-6)}{2} \right) \quad \checkmark \quad 1 \text{ mark}$$

$$= \left(\frac{4}{2}, \frac{-4}{2} \right)$$

$$= (2, -2)$$

(c) Gradient AC

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-6 - 2}{6 - (-2)}$$

$$= \frac{-8}{8}$$

$$\therefore m = -1 \quad \checkmark \quad 1 \text{ mark}$$

$$(d) d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(+6 - (-2))^2 + (-6 - 2)^2}$$

$$= \sqrt{64 + 64}$$

$$= \sqrt{128} \quad \checkmark \quad 1 \text{ mark}$$

$$= 8\sqrt{2}$$

(e) $m_{AC} = -1$ (from c)

$$m_{BP} = \frac{y_2 - y_1}{x_2 - x_1} \quad B(-1, -5) \quad P(2, -2)$$

$$= \frac{-2 + 5}{2 + 1}$$

$$= \frac{3}{3}$$

$$= 1$$

$$= 1 \quad \checkmark$$

Now $m_{AC} \times m_{BP} = -1 \times 1$
 $= -1$ $\checkmark$

$\therefore AC \perp BP$

2marks.

(f) $A = \frac{1}{2}bh$

$$= \frac{1}{2} \times AC \times BP$$

$$= \frac{1}{2} \times 8\sqrt{2} \times \sqrt{18}$$

$$= 4\sqrt{36}$$

$$= 24 \text{ units}^2 \quad \checkmark$$

$$BP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(2+1)^2 + (-2+5)^2}$$

$$= \sqrt{9+9}$$

$$= \sqrt{18} \quad \checkmark$$

2marks

(g) line AC $A(-2, 2) \quad m = -1$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -1(x + 2) \quad \checkmark$$

$$y - 2 = -x - 2$$

$$y = -x$$

$$x + y = 0$$

2marks.